Project Initialization and Planning Phase

Date	28 June 2026
Team ID	SWTID-2026-9346
Project Title	Voice--Based--Concept--Understanding--Analyzer
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview

Category	Details
Objective	The primary objective is to develop an AI-powered system that evaluates a user's conceptual understanding by analyzing their spoken explanation. The system provides accurate, objective, and personalized feedback using speech recognition, semantic analysis, and AI-generated insights.
Scope	The project covers voice recording/upload, speech-to-text conversion, semantic similarity analysis, speech feature extraction, AI-based feedback generation, performance scoring, waveform visualization, PDF report generation, and storage of evaluation history for future reference.

Problem Statement

Category	Details
Description	Traditional methods of evaluating conceptual understanding, such as viva examinations and oral presentations, are manual, time-consuming, subjective, and often inconsistent. Existing speech recognition systems convert speech into text but do not evaluate the quality or correctness of the explanation.
Impact	Solving this problem enables faster, objective, and consistent evaluation of conceptual understanding, reduces teachers' evaluation workload, improves students' learning outcomes, and provides personalized feedback for continuous improvement.

Proposed Solution

Category	Details
Approach	Develop an AI-powered Voice-Based Concept Understanding Analyser that uses Whisper for speech-to-text conversion, Sentence-BERT for semantic similarity analysis, Librosa for speech feature extraction, and Google Gemini AI for personalized feedback and recommendations.
Key Features	• Voice recording and audio upload.• Automatic speech-to-text transcription using Whisper.• Semantic similarity analysis using Sentence-BERT.• Speech fluency and audio feature analysis using Librosa.• AI-generated feedback and improvement suggestions using Gemini.• Performance scoring and visualization dashboard.• Downloadable PDF evaluation report.• Evaluation history and report storage.

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU/GPU for AI model inference and processing	Intel Core i5/i7 or AMD Ryzen 5+, NVIDIA T4 GPU (optional for faster processing)
Memory	RAM required for AI libraries and processing	Minimum **8 GB RAM** (16 GB Recommended)
Storage	Space for source code, AI models, reports, and database	Minimum **100 GB SSD**

Software

Resource Type	Description	Specification / Allocation
Frameworks	Frontend and Backend Frameworks	React.js, Vite, FastAPI
Libraries	AI, Audio Processing, Visualization	OpenAI Whisper, Sentence-Transformers (SBERT), Librosa, NumPy, Pandas, Matplotlib, ReportLab, SoundFile
Development Environment	IDE and Development Tools	Visual Studio Code, PyCharm, Jupyter Notebook

Data

Resource Type	Description	Specification / Allocation
Data	Audio recordings, expected concept answers, transcripts, evaluation reports	User-recorded audio (.wav/.mp3), custom concept dataset, generated transcripts (TXT/JSON), SQLite/PostgreSQL database